- *Physical context* describes how components combine into larger systems. A digital twin models the hierarchy from individual parts up to complete assemblies, for example, a temperature sensor is part of an HVAC unit, which is part of a room, which is part of a building. This mirrors real-world structure rather than arbitrary data organization.

Figure 5.4 shows how these contexts work together for a garden moisture sensor. The sensor produces a reading of 15% —raw data without meaning. Temporal context shows moisture has dropped from 35% over three days. Spatial context places the sensor in the southeast corner, three meters from Sprinkler A and eight meters from Sprinkler B. Physical context reveals Sprinkler A connects to Solenoid 2. Relational context groups the sensor and Sprinkler A into "Zone 1" with shared watering rules.

Together, these contexts enable the digital twin to reason: "Zone 1 moisture is critically low and dropping. Activate Solenoid 2 to run Sprinkler A for 20 minutes". Without this contextual understanding, the digital twin sees only disconnected facts like a 15% reading here, and a solenoid identifier there, with no ability to take meaningful action.

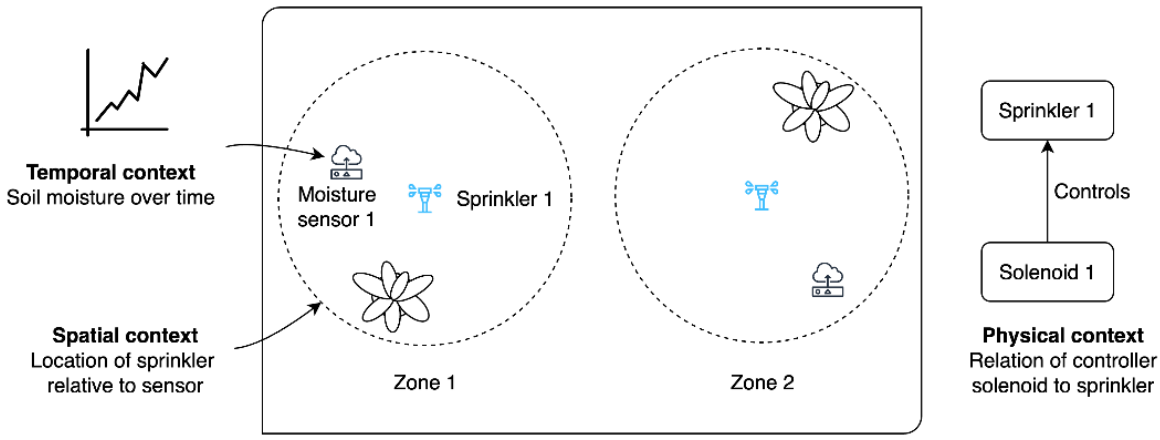


Figure 5.4 An example of how raw data produced by a moisture sensor in my garden can be contextualized with temporal, spatial, physical, and relationship context.

5.2.2 How to contextualize data

Unless you are starting from scratch, building a model of a physical system normally means that you will likely be dealing with many digital representations of the same thing. William Kent's 1978 book *Data and Reality*